

# TO YIN YU

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## EDUCATION

### University of Maryland, College Park — College Park, MD

Bachelor of Science, Computer Science · December 2022

**Relevant coursework:** Advanced Data Structures, Artificial Intelligence, Data Science, Computer Architecture, Object-Oriented Programming, Functional Programming, Web Development, Computer Systems, Programming Handheld Systems.

## TECHNICAL SKILLS

**Languages:** Python, JavaScript/TypeScript, Java, Swift, C, Rust, OCaml, Ruby, SQL

**ML & Data:** TensorFlow, Keras, scikit-learn, Pandas, NumPy, matplotlib, seaborn, SAS, Minitab

**Web & Backend:** React, Next.js, Astro, FastAPI, Tailwind CSS, PostgreSQL, Redis

**Tools:** Docker, Git, Linux/Unix, Vercel, Netlify, Jupyter

## PROJECTS

### SmartUrl — URL Shortening Platform | Python, FastAPI, PostgreSQL, Redis, Docker

[github.com/tonyx1998/SmartUrlServices](https://github.com/tonyx1998/SmartUrlServices)

- Built a full-stack URL shortener with QR and barcode generation, click and scan analytics, and automatic page-title extraction.
- Implemented Redis-backed three-tier rate limiting (10 / 60 / unlimited req/min) and response caching to cut hot-path latency.
- Containerized every service with Docker Compose for reproducible local development and production deployments.

### United Front Roofing — Marketing Site | Astro, Tailwind CSS, JavaScript, Netlify

[united-front-roofing.netlify.app](https://united-front-roofing.netlify.app)

- Designed and shipped a complete marketing site for a local roofing business — landing pages, project gallery, and SEO-optimized routing.
- Built a rule-based AI chat assistant (no third-party API), a multi-step quote estimator, and a booking calendar.

### Personal Portfolio | Next.js 16, TypeScript, Tailwind CSS v4, Framer Motion

[personal-site-liart-beta.vercel.app](https://personal-site-liart-beta.vercel.app)

- Built a responsive portfolio with dark/light theming, scroll-driven animations, a bento-grid project layout, and cursor-tracked spotlight cards.
- Implemented an in-house, React-19-clean theme provider with FOUC-prevention via a *beforeInteractive* inline script and a working accessible Contact form backed by Web3Forms.

### Obesity Data Analysis | Python, Pandas, NumPy, scikit-learn, matplotlib, seaborn

[tonyx1998.github.io](https://tonyx1998.github.io)

- Analyzed obesity patterns from public-health datasets, surfacing demographic and lifestyle trends with matplotlib/seaborn visualizations.
- Built and validated predictive models with scikit-learn and statsmodels to identify statistically significant risk factors.

### Schedule Creator | Java

[github.com/tonyx1998/Schedule-Creator](https://github.com/tonyx1998/Schedule-Creator)

- Built a Java desktop application that generates conflict-free class schedules by modeling courses, time slots, and scheduling constraints.